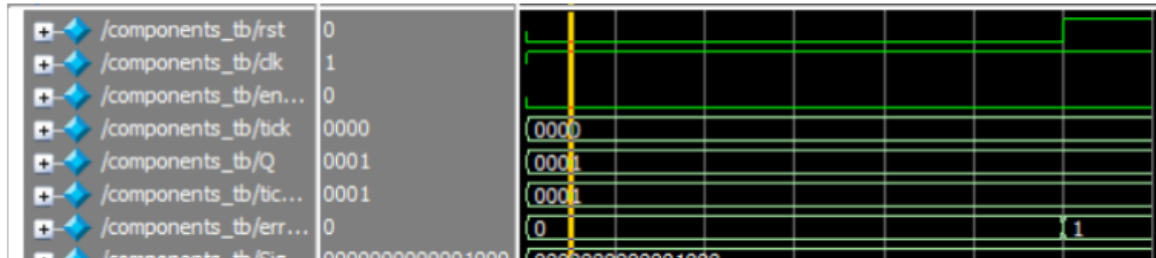


Demonstration Plan

Design of Components & Test Plans:

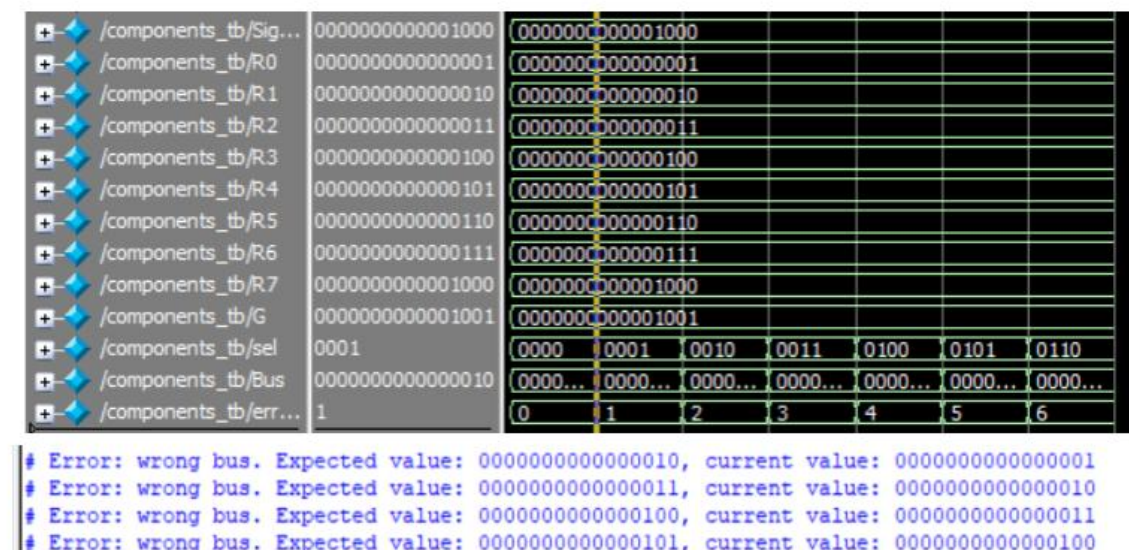
Debugging for component testbench.

tick_FSM



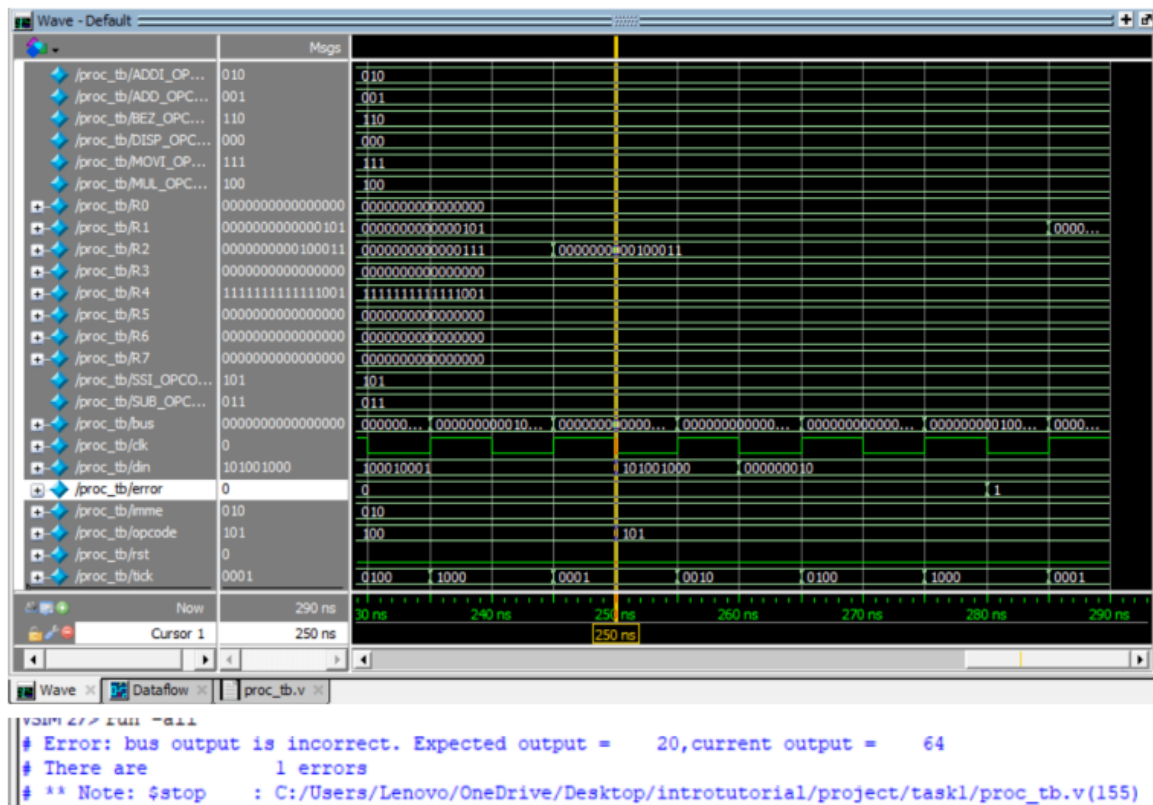
It is realised that 'enabled' was not updated and always remains a 0, therefore the whole test remains in the same state.

Multiplexer



The bus was technically correct in the graph, however the test sensed the previous Bus value for subsequent cases of sel values after 10ns.

For task 2/3:



When testing for ssi operation, the test shows that the module did not reach the expected answer. The time that starts checking for ssi operation is at the yellow line, at 250 ns.

Live Hardware Demonstration Plans:

For proc, do the following to illustrate the results:

1. MOVI R1, 5 – Loads 5 into R1.
2. ADDI R2, 2 – Adds 2 to R2.
3. ADD R2, R1 – Adds R1 to R2.
4. SUB R4, R2 – Subtracts R2 from R4.
5. MUL R2, R1 – Multiplies R2 and R1.
6. SSI R1, 2 – Shifts R1 left by 2 bits.

For proc_extension, do the following to illustrate the results:

1. MOVI R1, 31 – Loads 31 into R1.
2. DISP R1 – Displays R1.
3. ADDI R2, 63 – Adds 63 to R2.
4. DISP R2 – Displays R2.
5. SUB R1, R2 – Subtracts R2 from R1.
6. DISP R1 – Displays the final R1 value.

Live Testbench Demonstration Plans:

For proc_tb module:

The testbench demonstrates the basic arithmetic and logic operations of the processor.

1. MOVI R1, 5 – Loads 5 into R1.
2. ADDI R2, 2 – Adds 2 to R2.
3. ADD R2, R1 – Adds R1 to R2.
4. SUB R4, R2 – Subtracts R2 from R4.
5. MUL R2, R1 – Multiplies R2 and R1.
6. SSI R1, 2 – Shifts R1 left by 2 bits.

For proc_extension_tb module:

This extended testbench includes all the above operations and also verifies the display output functionality.

1. MOVI R1, 5 – Loads 5 into R1.
2. ADDI R2, 2 – Adds 2 to R2.
3. ADD R2, R1 – Adds R1 to R2.
4. SUB R4, R2 – Subtracts R2 from R4.
5. MOVI R3, 2 – Loads 2 into R3.
6. MOVI R5, 8 – Loads 8 into R5.
7. SUB R3, R5 – Subtracts R5 from R3 (expected result = -6).
8. DISP R3 – Displays R3 (negative result).
9. MUL R2, R1 – Multiplies R2 and R1.
10. SSI R1, 2 – Shifts R1 left by 2 bits.
11. DISP R1 – Displays R1.
12. MOVI R6, 120 – Loads 120 into R6.
13. ADDI R6, 8 – Adds 8 to R6 (R6 = 128).
14. MOVI R7, 50 – Loads 50 into R7.
15. MUL R6, R7 – Multiplies R6 and R7 (expected result = 6400).
16. DISP R6 – Displays R6 (large multiplication result).
17. MOVI R0, 0 – Loads 0 into R0.
18. ADD R1, R0 – Adds R0 to R1 (R1 unchanged).
19. DISP R1 – Displays R1.
20. SUB R2, R0 – Subtracts R0 from R2 (R2 unchanged).
21. DISP R2 – Displays R2.
22. MOVI R3, 10 – Loads 10 into R3.
23. MOVI R4, 2000 – Loads 2000 into R4 (symbolic/placeholder).
24. SUB R3, R4 – Subtracts R4 from R3 (expected result $\approx$ -2000).
25. DISP R3 – Displays R3 (large negative result).

This confirms that the processor's control unit, ALU, registers, and output display are functioning correctly together.